COMP30027 Project 2 Report

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1. Introduction

The project applies a coarse-to-fine approach for image classification. This approach first involves grouping instances into broader categories, and then using some of this information to categorise into more specific ones. Therefore, this project will be split into two tasks. The first task is responsible for classifying images into 10 broad animal categories, and figuring out what factors contribute to reasonable performance. The second task applies all the learnings from Task 1, and outlines if performance has improved or not to attempt fine-grained classification of bird species. The report outlined that performance worsened during the transition from coarse-to-fine. While using ResNet and other convolutional neural networks as feature extractors transfers well from coarse-to-fine grained classification, using MI and handcrafted features under a Random Forest model did not work as well.

2. Method

2.1 Feature Engineering

The provided features include colour information, which provides a feature vector of binned data. HOG features, which capture the structure and shape of an object and has PCA applied to it. There are also additional features which include edge density, texture variance and average colour channels.

As well as utilising provided features, I have also engineered my own features. The engineered features include edge detection, which tries to identify boundaries and specific parts of an image where the pixel blindness changes sharply[1]. This helps to identify anatomical shapes and morphological features.

Another feature that was engineered was SIFT features which aim to describe distinctive spatial locations. Descriptors were clustered into 100 groups using K-means. BoVW is then used to build more robust image embeddings. This amplifies SIFT's property of scale and rotational invariance, helping to recognise images at different sizes and orientations.

Texture features have also been extracted, which aims to describe the tactile quality of an image. Raw images are converted into uniform standardised pixels in byte format which gets processed into a Local Binary Pattern

These features are appropriate for coarse-grained classification because it captures broad, structural patterns and are useful when animal images are distinct from each other.

2.2 Task 1

This task will investigate how model and feature selection impact coarse-grained image classification.

An SVM model is trained on features extracted from a pre-trained ResNet-50 model, which is a type of convolutional neural network. SVM models are appropriate for image classification tasks due to its ability to handle high dimensional data as it only focuses on support vectors closest to the hyperplane. To generate the feature dataset, images from the training set are re-sized into 224 x 224 dimensions, which matches the dimensions of ImageNet instances and are normalised with the mean and standard deviation of ImageNet values. The new feature set is split into a training and validation set. Using stratified cross-validation, SVM hyperparameters are tuned and performance is evaluated using Holdout Validation. This approach ensures that the hyperplane can be focused on important features from the image dataset, filtering out noise and ensures that it is easier to maximise the optimal decision boundary of the hyperplane.

The engineered and provided features will be combined for the intention of calculating mutual information scores, which can identify good features. For a feature to be considered good, it must be highly correlated and have the most influence on the target variable. The number of features based on mutual information will be a hyperparameter and will be determined by stratified cross-validation. A Random Forest model will be trained on these features. Random Forest is an ensemble machine learning model that aims to combine the results of multiple decision trees to provide a meaningful output. Using mutual information is appropriate as a metric due to its ability to high-dimensional, non-linear data which is present in images. To determine if model selection is important for coarse-grained classification, an SVM model will also be trained on those handcrafted features after performing MI selection, for comparison purposes.

2.1 Task 2

The second task will train the SVM + ResNet and the Random Forest + Hand-engineered models on the datasets for fine-grained bird classification. The intention is to replicate the earlier task as much as possible to characterise the coarse-to-fine progression. The MI scores for the engineered features will be recomputed to investigate if feature relevance varied across both tasks.

Considering that dataset in Task 2 was smaller, stratified cross-validation was used as an evaluation metric instead of holdout validation because it provides a more reliable estimate of a model's performance than holdout validation in smaller datasets.

Extending upon the findings of Task 1, a bagging ensemble method is used to potentially identify alternative models that can improve fine-grained classification. It is an ensemble learning method that trains multiple classifiers and then combines their predictions. For the training of classifiers, two types of feature extractors have been used, ResNet-50 and EfficientNet, which is another convolutional neural network that slightly improves from ResNet[2]. Logistic regression is used as a base classifier. This is appropriate for fine-grained image classification because this method will ensure that the model because it creates a more robust model that aims to leverage different architectural strengths.

3. Results

SVM – Support Vector Machine
RF – Random Forest
HC – Handcrafted + Provided Features

Model	SVM + ResNet	RF + HC Features	SVM + HC Features
Validation	0.915	0.567	0.376
50% Kaggle Test	0.929	0.569	0.384

Table 1- Accuracy Scores of Validation and 50% Kaggle Test Sets for Task 1 models

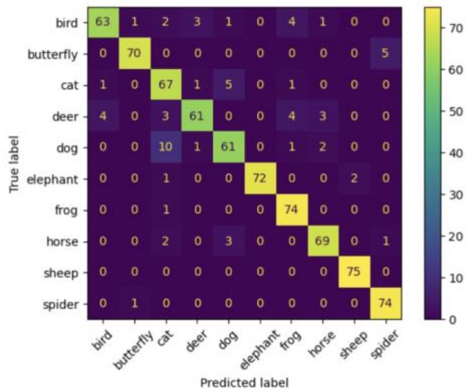

Figure 1- Confusion Matrix for the SVM + ResNet model for Task 1 from the Validation Set

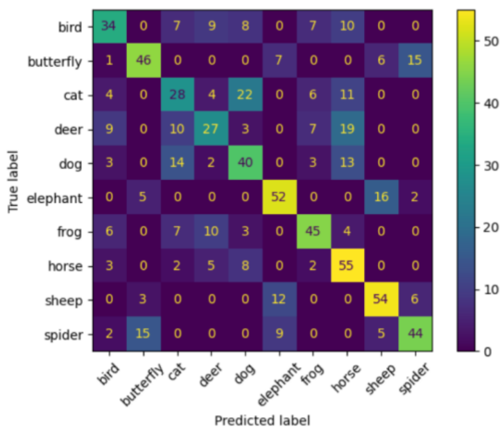

Figure 2- Confusion Matrix of Random Forest Model with a mixture of provided and handcrafted features for Task 1 from the Validation Set

Figure 3- Subset of cat instances misclassified as dogs from the Random Forest model in Task 1

Feature	MI Score
texture_0	0.760
texture_9	0.712

feat_1	0.697
texture_7	0.678
texture_1	0.650
texture_4	0.620
texture_5	0.610
texture_3	0.545
texture_8	0.442
texture_2	0.255
feat_0	0.242
edge_11	0.143
texture_6	0.109
color_31	0.105
color_95	0.102
feat_8	0.100
edge_1	0.098
edge_10	0.091
color_62	0.089
color_34	0.088

Table 2- Top 20 features by MI Score for Task 1 with Provided and Handcrafted features combined

Class	SVM + ResNet	RF + HC
Bird	0.881	0.496
Butterfly	0.952	0.639
Cat	0.832	0.392
Deer	0.865	0.409
Dog	0.841	0.503
Elephant	0.980	0.671
Frog	0.931	0.621
Horse	0.920	0.588
Sheep	0.987	0.692
Spider	0.955	0.620

Table 3- Class-level F1- scores for models in Task 1

Class	SVM + ResNet	RF + HC
Bird	0.840	0.453
Butterfly	0.933	0.613
Cat	0.893	0.373
Deer	0.813	0.360
Dog	0.813	0.533
Elephant	0.960	0.693
Frog	0.987	0.600
Horse	0.920	0.733
Sheep	1.000	0.720
Spider	0.987	0.587

Table 4- Class-level Recall for models in Task 1

Class	SVM + ResNet	RF + HC
Bird	0.926	0.548
Butterfly	0.972	0.667
Cat	0.779	0.412
Deer	0.924	0.474
Dog	0.871	0.476
Elephant	1.000	0.650
Frog	0.881	0.643
Horse	0.920	0.491
Sheep	0.974	0.667
Spider	0.925	0.657

Table 5- Class-level Precision scores for models in Task 1

Model Architecture	Task 2: Validation Set	Task 2: 50% of Kaggle Test Set
SVM + ResNet Features	0.887	0.878
Random Forest + Handcrafted and Provided Features	0.333	0.378
Bagging + Logistic regression model	0.916	0.922

Table 6- Accuracy Scores of Validation and 50% Kaggle Test Sets for Task 1 models

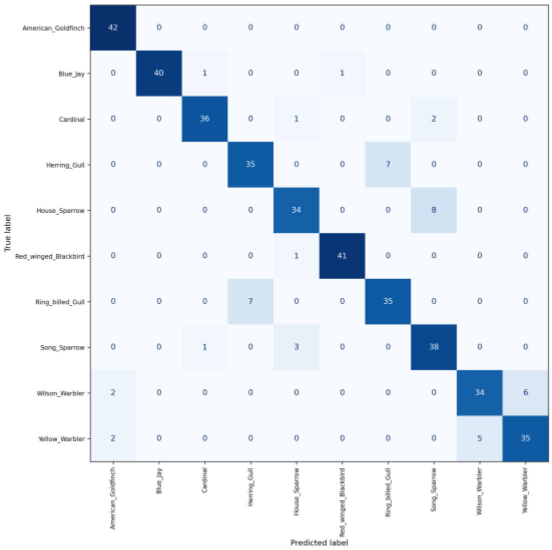

Figure 4- Confusion Matrix for the SVM + ResNet model for Task 2 from the Validation Set

Figure 5- Subset of House Sparrows instances misclassified as Song Sparrows from the SVM Model in Task 2

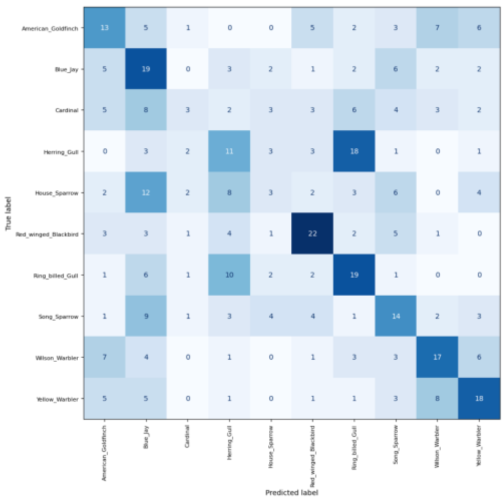

Figure 6- Confusion Matrix for Random Forest model for Task 2 from the Validation Set (for one specific run of Random Forest)

Figure 7- Subset of Herring Gull instances misclassified as Ring-billed Gull from the Random Forest Model in Task 2 (for one specific run of Random Forest)

Figure 8- Subset of House Sparrow instances misclassified as Blue Jays from the Random Forest Model in Task 2 (for one specific run of Random Forest)

Rank	Feature Name	MI Score
1	hog_pca_0	0.208
2	edge_2	0.161
3	edge_10	0.159
4	edge_7	0.157
5	edge_6	0.149
6	feat_7	0.144
7	color_36	0.139
8	color_73	0.137
9	edge_3	0.134
10	texture_9	0.133
11	edge_0	0.124
12	sift_18	0.123
13	color_68	0.117
14	edge_4	0.117
15	texture_2	0.114
16	color_2	0.108
17	color_45	0.107
18	color_44	0.107
19	color_57	0.106
20	color_6	0.104

Table 7- Accuracy Scores of Validation and 50% Kaggle Test Sets for Task 2 models

Model	SVM + ResNet	Random Forest + HC
American_Goldfinch	0.955	0.310
Blue_Jay	0.976	0.328
Cardinal	0.935	0.120

Herring_Gull	0.833	0.259
House_Sparrow	0.840	0.100
Red_Winged_Blackbird	0.976	0.512
Ring_billed_Gull	0.833	0.384
Song_Sparrow	0.844	0.318
Wilson_Warbler	0.840	0.415
Yellow_Warbler	0.843	0.429

Table 8- Class-level F1- scores for models in Task 2 (for one specific run for Random Forest)

Model	SVM + ResNet	Random Forest + HC
American_Goldfinch	1.000	0.310
Blue_Jay	0.952	0.452
Cardinal	0.923	0.077
Herring_Gull	0.833	0.262
House_Sparrow	0.810	0.071
Red_Winged_Blackbird	0.976	0.524
Ring_billed_Gull	0.833	0.452
Song_Sparrow	0.905	0.333
Wilson_Warbler	0.810	0.405
Yellow_Warbler	0.833	0.429

Table 9- Class-level Recall scores for models in Task 2 (for one specific run for Random Forest)

Model	SVM + ResNet	Random Forest + HC
American_Goldfinch	0.913	0.310
Blue_Jay	1.000	0.257
Cardinal	0.947	0.273
Herring_Gull	0.833	0.256
House_Sparrow	0.872	0.167
Red_Winged_Blackbird	0.976	0.500
Ring_billed_Gull	0.833	0.333
Song_Sparrow	0.792	0.304
Wilson_Warbler	0.872	0.425
Yellow_Warbler	0.854	0.429

Table 8- Class-level Precision scores for models in Task 2 (for one specific run for Random Forest)

4. Discussion

Considering that the baseline accuracy for a 10-class classification problem is 10%, as the accuracy levels across all models in both tasks exceeded this, this means that these models actually attempted to learn.

A key finding in Task 1 was that selecting an ideal feature representation contributed more to performance, rather than model selection. When comparing accuracy scores amongst the Kaggle data in Table 1, there is a larger gap in performance (54.5%) when feature representation changes, compared to when the model changes (19.7%). For coarse-grained animal classification, classes are visually distinct from each other. A strong feature extractor would be able to meaningfully separate the classes, which means that decision boundaries can be easier to define, even in simpler models.

Pre-trained features on a ResNet-50 model generally performed better than handcrafted features such as edges, SIFT and textures, when comparing accuracies of both SVM models in Task 1. The Image-Net dataset that the ResNet model trains on contains millions of images. This gives it the ability to recognise hierarchical features through gradually learning more complex features in each layer. This exposes a limitation of using handcrafted features as it introduces human-induced bias, rather than directly learning a representation from the data.

According to the confusion matrix at Figure 2, for the Random Forest model in Task 1, the most frequent misclassification was predicting cats as dogs which happened 22 times. As some of the misclassified images were zoomed in, since dogs and cats have similar fur textures, the model could not have meaningfully used these features to separate the two classes. The images in Figure 3, show the triangular pinna of cats, which can resemble dog breeds with a pointed ear structure[3]. This is a morphological similarity that hand-crafted features may not be able to distinguish.

However, when comparing the performance of Random Forest and SVM with engineered features in Table 1, Random Forest performed better. Multiclass classification is achieved with SVM using One vs Rest, where multiple classifiers are trained with different classes being a target, and then classifying based on the highest confidence score of individual classifiers. While each class is trying to find the optimal hyperplane that separates from the rest, Random Forest can capture more localised interactions between features through using multiple decision trees on different splits of the data, making it perform better for coarse-grained animal classification.

The top 20 MI features change across both tasks in Table 2 and Table 7. Most top-ranked features for coarse-grained animal classification are texture-based features which are more discriminative than colour and edge-based features. Textures capture the main properties of an animal's surface such as fur and feathers. Conversely, many animals share colours (for example both horses and deer are brown). Edge details are also lost at a 64 x 64 size with lower resolutions. However, these rankings did not transfer onto fine-grained classification as HOG and edge-features dominated instead. These features are good at capturing structural attributes such as beak shape and wing contours. However, the top 20 features had lower MI score in the fine-grained version on average, implying that association between handcrafted features and class labels are weaker.

Considering the coarse-to-fine progression, the validation accuracy of the SVM model slightly decreased from 0.915 to 0.887. Interestingly, unlike the SVM model in Task 1, the validation accuracy is lower than the test accuracy in Table 6 due to overfitting introduced from cross-validation, where each fold contained fewer samples.

According to Figure 4, the model most frequently confused House sparrows as Song Sparrows 8 times. Considering that the SVM model was tuned with an RBF Kernel with higher flexibility and a higher penalty for mistakes with C = 10, this suggests that it struggled to discriminate between the two species. Further examination of these misclassified instances suggests that both species share brown plumage and body proportions. Additionally, when House sparrows angle their wings, it resembles the dark breast streaks that Song sparrows have.

The decrease in accuracy demonstrates inherent limitations when attempting fine-grained classification. ResNet struggles on features that are harder to discriminate against, which are more prevalent in fine-grained classifications, meaning that ResNet works better in Task 1 than in Task 2.

Nevertheless, the high F1-scores amongst all the bird species in Table 8 suggest that ResNet-50 extracted features built on an SVM model transferred well from coarse-to-fine grained classification.

The same cannot be said for the Random Forest model with engineered features, where the performance significantly dropped during fine-grained image classification. Every class had an F1-score below 0.6 in the validation set, according to Table 8, implying that the engineered features may not contain enough information to separate the classes.

The model seems to make two types of mistakes. Looking at the confusion matrix, the most confused classes at Figure 6, Herring Gulls were being predicted as Ring-billed gulls 18 times, while House Sparrows were being predicted as Blue jay 12 times. The model both struggled when bird species are difficult to discriminate as well as struggling when the two species are visually distinct. The dominance of HOG and edge features doesn't provide much discriminative power, as Herring gulls and Ring-billed gulls only differ by subtle bill differences. However, the confusion between House Sparrows and Blue jays is interesting considering colour can be used to distinguish between the two species interestingly and colour features appear frequently in the Top 20 MI features in Table 9. However, the colour features ranked lower than the edge features. When looking at the misclassified images, House Sparrows and Blue jays share similar size and morphological structure, meaning that the model was not able to capture obvious differences.

Therefore, using Random Forest with a mix of handcrafted and provided features does not transfer well because these handcrafted features were not able to meaningfully discriminate between bird species with less distinct differences.

The coarse-to-fine grain approach exposes some key limitations. MI-based feature selection is not as effective if the top-ranking features do not meaningfully discriminate between classes themselves. Rich feature extractors have a ceiling when it comes to separating classes, as it struggled when bird species looked nearly identical. Additionally, the using handcrafted features and performing manual feature engineering has trouble forming deep, hierarchical associations, unlike models like neural networks. Additionally, the smaller-sized dataset in Task 2 compounded this.

When considering alternative models, classifier combination models can be considered as an alternate model for fine-grained classification. These work well when multiple classifiers can make different types of mistakes, because when independent errors are made by different models, these errors can be cancelled out through aggregation. As seen in Table 6, the bagging model outperformed the others. EfficientNet and ResNet-50 are both convolutional neural networks, which can already guarantee reasonable performance as feature extractors as observed above. Both models have different architectural approaches, as ResNet focuses on high-level structural features while EfficientNet focuses on fine-grained pattern recognition. Furthermore, as bagging can reduce variance, without introducing more bias, this would have led to better performed amongst fine-grained classification.

5. Conclusion

The SVM + ResNet model was the best performing model out of models involved in both tasks, demonstrating that different granularity levels can use deep feature extraction. As performance declined from Task 1 to Task 2 on average, this suggests that the coarse-to-fine approach is a difficult process to get correct as both tasks require different approaches when separating classes. The main factors to consider is how to separate visually similar classes and how to select a correct feature extraction approach. The small dataset, reflecting real world constraints may have contributed further to the drop in performance.

6. References

[1]:**https://www.geeksforgeeks.org/machine-learning/image-feature-extraction-using-python/**

[2]:https://medium.com/@enrico.randellini/image-classification-resnet-vs-efficientnet-vs-efficientnet-v2-vs-compact-convolutional-c205838bbf49

[3]:**https://www.oakleighcentralvet.com.au/single-post/anatomy-of-the-ear-cats-and-dogs**

7. Word Count

Introduction(130) + Method(700) + Results(12) + Discussion(1166) + Conclusion(97) = 2105 words